



December 6, 2016

NG-16-0226  
10 CFR 50.73

U.S. Nuclear Regulatory Commission  
Attn: Document Control Desk  
Washington, D.C. 20555-0001

Duane Arnold Energy Center  
Docket 50-331  
Renewed Op. License No. DPR-49

Licensee Event Report 2016-002

Please find attached the subject report submitted in accordance with 10 CFR 50.73. This letter makes no new commitments or changes to any existing commitments.

A handwritten signature in cursive script, appearing to read "Steven P. Brown".

*Fory* T. A. Vehec  
Vice President, Duane Arnold Energy Center  
NextEra Energy Duane Arnold, LLC

cc: Administrator, Region III, USNRC  
Project Manager, DAEC, USNRC  
Resident Inspector, DAEC, USNRC

IE22  
NRR

**LICENSEE EVENT REPORT (LER)**(See Page 2 for required number of  
digits/characters for each block)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

**1. FACILITY NAME**

Duane Arnold Energy Center

**2. DOCKET NUMBER**

05000-331

**3. PAGE**

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**4. TITLE**

Unplanned RCIC Inoperability Results in Safety System Functional Failure

| 5. EVENT DATE                                                                                        |     |      | 6. LER NUMBER                               |                   |         | 7. REPORT DATE                              |     |      | 8. OTHER FACILITIES INVOLVED                          |               |                                               |
|------------------------------------------------------------------------------------------------------|-----|------|---------------------------------------------|-------------------|---------|---------------------------------------------|-----|------|-------------------------------------------------------|---------------|-----------------------------------------------|
| MONTH                                                                                                | DAY | YEAR | YEAR                                        | SEQUENTIAL NUMBER | REV NO. | MONTH                                       | DAY | YEAR | FACILITY NAME                                         | DOCKET NUMBER |                                               |
| 4                                                                                                    | 28  | 2016 | 2016                                        | 002               | 00      | 12                                          | 6   | 2016 | N/A                                                   | N/A           |                                               |
| <b>9. OPERATING MODE</b>                                                                             |     |      |                                             |                   |         |                                             |     |      |                                                       |               |                                               |
| <b>11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)</b> |     |      |                                             |                   |         |                                             |     |      |                                                       |               |                                               |
| 1                                                                                                    |     |      | <input type="checkbox"/> 20.2201(b)         |                   |         | <input type="checkbox"/> 20.2203(a)(3)(i)   |     |      | <input type="checkbox"/> 50.73(a)(2)(i)(C)            |               | <input type="checkbox"/> 50.73(a)(2)(vii)     |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2201(d)         |                   |         | <input type="checkbox"/> 20.2203(a)(3)(ii)  |     |      | <input type="checkbox"/> 50.73(a)(2)(ii)(A)           |               | <input type="checkbox"/> 50.73(a)(2)(viii)(A) |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2203(a)(1)      |                   |         | <input type="checkbox"/> 20.2203(a)(4)      |     |      | <input type="checkbox"/> 50.73(a)(2)(ii)(B)           |               | <input type="checkbox"/> 50.73(a)(2)(viii)(B) |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2203(a)(2)(i)   |                   |         | <input type="checkbox"/> 50.36(c)(1)(i)(A)  |     |      | <input type="checkbox"/> 50.73(a)(2)(iii)             |               | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   |
| 10. POWER LEVEL<br><br>100%                                                                          |     |      | <input type="checkbox"/> 20.2203(a)(2)(ii)  |                   |         | <input type="checkbox"/> 50.36(c)(1)(ii)(A) |     |      | <input type="checkbox"/> 50.73(a)(2)(iv)(A)           |               | <input type="checkbox"/> 50.73(a)(2)(x)       |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2203(a)(2)(iii) |                   |         | <input type="checkbox"/> 50.36(c)(2)        |     |      | <input type="checkbox"/> 50.73(a)(2)(v)(A)            |               | <input type="checkbox"/> 73.71(a)(4)          |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2203(a)(2)(iv)  |                   |         | <input type="checkbox"/> 50.46(a)(3)(ii)    |     |      | <input type="checkbox"/> 50.73(a)(2)(v)(B)            |               | <input type="checkbox"/> 73.71(a)(5)          |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2203(a)(2)(v)   |                   |         | <input type="checkbox"/> 50.73(a)(2)(i)(A)  |     |      | <input type="checkbox"/> 50.73(a)(2)(v)(C)            |               | <input type="checkbox"/> OTHER                |
|                                                                                                      |     |      | <input type="checkbox"/> 20.2203(a)(2)(vi)  |                   |         | <input type="checkbox"/> 50.73(a)(2)(i)(B)  |     |      | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(D) |               | Specify in Abstract below or in NRC Form 366A |

**12. LICENSEE CONTACT FOR THIS LER****LICENSEE CONTACT**

Bob Murrell, Licensing Engineer

**TELEPHONE NUMBER (Include Area Code)**

(319) 851-7900

**13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT**

| CAUSE | SYSTEM | COMPONENT | MANU-FACTURER | REPORTABLE TO EPIX | CAUSE | SYSTEM | COMPONENT | MANU-FACTURER | REPORTABLE TO EPIX |
|-------|--------|-----------|---------------|--------------------|-------|--------|-----------|---------------|--------------------|
| A     | BN     | N/A       | N/A           | Y                  | N/A   | N/A    | N/A       | N/A           | N/A                |

**14. SUPPLEMENTAL REPORT EXPECTED**☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH DAY YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On April 28, 2016, at 1055, while operating at 100% power, with no structures, systems, or components inoperable that contributed to this event, during the performance of Surveillance Test Procedure (STP) 3.3.6.1-28, Reactor Core Isolation Cooling (RCIC) System Steam Line Flow – High Channel Functional Test, the RCIC turbine received an unplanned trip signal and subsequent turbine stop valve closure. The cause of the event was a human performance error associated with verifying the correct installation location of the relay block. The turbine trip resulted in the unplanned inoperability of RCIC, therefore, this condition meets the reporting requirements of 10CFR50.73(a)(2)(v)(D). At 1353 on April 28, 2016, the RCIC turbine was reset and RCIC was declared available. The safety significance of this event was low since all Emergency Core Cooling Systems were operable during the time the RCIC turbine was tripped.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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**NARRATIVE****I. Description of Event:**

On April 28, 2016, at 0848, while operating at 100% power, with no structures, systems, or components inoperable that contributed to this event, Surveillance Test Procedure (STP) 3.3.6.1-28, Reactor Core Isolation Cooling (RCIC) System (System Code BN) Steam Line Flow – High Channel Functional Test was started as part of planned testing. The purpose of STP 3.3.6.1-28 is to demonstrate operability of the Steam Line High Differential Pressure (flow) isolation instruments (PDIS-2441 and PDIS-2442) of the RCIC system. During the performance of the STP, relay blocks are installed to inhibit the RCIC turbine trip associated with the RCIC automatic isolation logic under test. Step 7.3.2, which functional checks PDIS-2442, required the installation of three relay blocks, two associated with the inboard isolation signals and one associated with the turbine trip. During the performance of subsequent steps, the RCIC turbine received an unexpected trip signal and MO-2405, RCIC Trip Throttle Valve closed and Control Room annunciator 1C04C (A-5) RCIC MO-2045 Turbine Trip actuated at 1055. Therefore, at 1055 on April 28, 2016, unplanned Limiting Condition for Operability (LCO), LCO 3.5.3 Condition A, RCIC System Inoperable, was entered. Investigations determined that the relay block for the PDIS-2442 turbine trip signal was placed on the contacts for PDIS-2441. At 1353 on April 28, 2016, the RCIC turbine was reset and RCIC was declared available and LCO 3.5.3 was exited.

**II. Assessment of Safety Consequences:**

The safety significance of this event was low. During the time that RCIC was unavailable, all of the Emergency Core Cooling Systems (High Pressure Coolant Injection, Core Spray, and Low Pressure Coolant Injection) that would have initiated in response to an accident or transient were operable.

All components and systems actuated as designed due to this event.

This event resulted in a Safety System Functional Failure and a Maintenance Preventable Functional Failure under the Maintenance Rule.

**III. Cause of Event:**

An Apparent Cause Evaluation (ACE) was completed for this event. The ACE determined that the direct cause of the RCIC turbine trip was the incorrect installation of relay block associated with inhibiting the RCIC turbine trip associated with the RCIC automatic isolation logic under test.

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## NARRATIVE

The Apparent Cause of the event was a human performance error associated with verifying the correct installation location of the relay block. Specifically, the use of Concurrent Verification was not consistent with procedural requirements. Contributing to this was inadequate placekeeping.

**IV. Corrective Actions:**Immediate Corrective Action

At 1353 on April 28, 2016, the RCIC turbine was reset and RCIC was declared available and LCO 3.5.3 was exited.

On May 6, 2016, a Human Performance Evaluation (HUPE) for the installation of the relay block on the incorrect contacts was completed.

On May 10, 2016, a Human Performance Department Clock Reset for the Instrument and Control Maintenance (ICM) department was completed to share lessons learned.

Corrective Actions for Cause of Event

The cause and contributing causes of this event were addressed by:

1. The implementation and tracking of the Management Action Response Checklist (MARC) process for the individuals involved with the installation of the relay in order to improve performance.
2. Development of Performance Focus Area for Training (PFAT) in ICM to ensure mastery of concurrent verification including proper placekeeping.
3. Development of a Dynamic Learning Activity for 'in cabinet' work to include expectations for concurrent verifications and placekeeping.

**V. Additional Information:**Previous Similar Occurrences:

A review of NextEra Energy Duane Arnold Licensee Event Reports from the previous three years found no other instances of events related to unplanned inoperability of RCIC.

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**NARRATIVE**

ELIS System and Component Codes:

BN - Reactor Core Isolation Cooling System

Reporting Requirements:

This activity is being reported pursuant to the requirements of 10CFR50.73(a)(2)(v)(D).